

Calibrating Temperature Sensors

PH3039 — Data Acquisition Lab Kenura R. Gunarathna | s17239 | Faculty of Science

Introduction and Setup

We calibrated an NTC thermistor using a **DS18B20** digital thermometer as the reference. The DS18B20 works from -55 to $+125$ °C, is accurate to ± 0.5 °C, and has a 9–12 bit resolution. Each one has its own 64-bit ID, so many sensors can share a single wire.

The thermistor and a fixed resistor form a voltage divider. A 10-bit ADC (0–1023 counts) reads the divider. We used three good runs: one heating and two cooling. The other runs had a stuck ADC channel. After cleaning, we had 1078 heating and 4241 cooling samples from 0 to 98 °C. We compare polynomial fits (degree 1–3) with the physical NTC β -model.

The DS18B20 uses the 1-Wire interface (one data pin plus ground) and works from 3.0 to 5.5 V. The experiment has two goals: to calibrate the thermistor, and to measure how fast it responds (its time constant).

Raw Data

Temperature against ADC always goes one way, but it is not a straight line. It is steeper at low counts. The time plots show a slow, smooth change in the water bath temperature, as expected for a thermal transient.

Heating and Cooling Calibration

Table 1. Data audit before fitting.

Item	Heating	Cooling
Samples logged	1079	4252
Empty (NaN) rows	1 (ADC 849)	2
Outliers removed ($> 2.5\sigma$)	0	9 (> 1.38 °C)
Samples used in fit	1078	4241
Files	141803	144604, 150749

We fitted polynomials of degree 1, 2 and 3 to temperature against ADC. For the cooling runs, we removed 2 empty rows and 9 outliers (more than 2.5 standard deviations, or 1.38 °C, from the fit).

Table 2. Fit errors (°C) for heating (H, $N = 1078$) and cooling (C, $N = 4241$).

Deg.	R^2 (H)	RMSE (H)	Max (H)	R^2 (C)	RMSE (C)	Max (C)
1	0.99600	2.326	3.913	0.99266	1.461	4.004
2	0.99718	1.951	4.146	0.99883	0.584	2.373
3	0.99991	0.345	1.459	0.99898	0.545	1.445

The cubic fit is the best. The straight line leaves an S-shaped error of up to about 4 °C. The cubic cuts the RMSE to 0.35 °C for heating and 0.55 °C for cooling. All cubic terms are significant ($|t| > 18$, $p < 0.001$). The large condition number ($\sim 10^9$) only comes from using raw ADC powers, so it is not a real problem. The cooling errors are strongly correlated in time (Durbin–Watson 0.02), so the quoted standard errors are a little too small.

Table 3. Cubic fit coefficients. All terms have $p < 0.001$.

Coefficient	Heating	Cooling	Global	Global std. err.
c_0 (intercept)	−18.792	−8.433	−17.558	0.076
c_1 (ADC)	0.2277	0.1562	0.2134	0.001
c_2 (ADC ²)	-3×10^{-4}	-1×10^{-4}	-2.495×10^{-4}	1.34×10^{-6}
c_3 (ADC ³)	2.107×10^{-7}	1.186×10^{-7}	1.869×10^{-7}	9.36×10^{-10}
N / AIC	1078 / 773.8	4250 / 7028	5319 / 8544	—

Table 4. Residual checks for the cubic fits.

Statistic	Heating	Cooling	Global
F -statistic	4.06×10^6	1.35×10^6	3.22×10^6
Durbin–Watson	0.156	0.022	0.031
Jarque–Bera	7.80	321.1	466.7
Skew / kurtosis	0.21 / 2.98	−0.62 / 2.48	−0.72 / 2.80
Condition number	2.6×10^9	1.3×10^{10}	2.4×10^9

The heating residuals are close to normal (skew 0.2, kurtosis 3). The cooling and global residuals are slightly skewed. This means the fit has a small leftover shape error, so the 95 % band is a rough guide, not an exact limit.

Heating vs. Cooling Hysteresis

We compared the two cubic fits over the shared range (ADC 309–849). The biggest gap is **1.16 °C** (1.16 % of full scale) and the average gap is **0.61 °C** (Figure 1). There are three likely causes:

- 1. Thermal lag.** The bead and the reference probe have different time constants $\tau = R_{th} C_{th}$, so one sensor trails the other while the temperature changes.
- 2. Maths effect.** The gap ΔT is the difference of two cubics, so it is always a smooth wave that crosses zero (here near ADC ≈ 500).
- 3. Water bath.** Without good stirring, warm water rises and the two probes see different temperatures.

The average gap is smaller than the reference uncertainty, so we do not claim real sensor hysteresis.

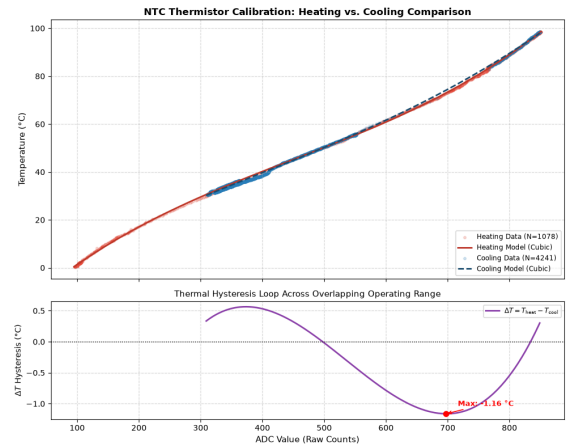


Figure 1. Heating vs. cooling fits and the gap ΔT .

Global Calibration and 95 % Prediction Band

We joined the heating data and the cleaned cooling data ($N = 5319$) and fitted one cubic, where a is the ADC count:

$$T = 1.869 \times 10^{-7} a^3 - 2.495 \times 10^{-4} a^2 + 0.2134 a - 17.558 \quad (1)$$

It has $R^2 = 0.999$ and RMSE 0.54 °C. The mean 95 % prediction half-width is **± 1.06 °C** (Figure 2).

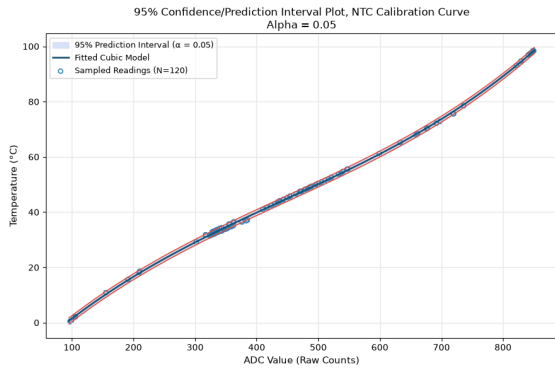


Figure 2. Global cubic fit with its 95 % prediction band.

Theoretical NTC β -Model

The divider output is $V_{out} = V_{cc} R_f / (R_{NTC} + R_f)$. The ADC uses V_{cc} as its reference, so the reading does not change if the supply drifts:

$$ADC = ADC_{max} \cdot \frac{R_f}{R_{NTC} + R_f} \Rightarrow r \equiv \frac{R_{NTC}}{R_f} = \frac{1023 - ADC}{ADC} \quad (2)$$

Rearranging $ADC (R_{NTC} + R_f) = ADC_{max} R_f$ gives $ADC \cdot R_{NTC} = (ADC_{max} - ADC) R_f$. Dividing by $ADC \cdot R_f$ gives the ratio r above. Here $ADC_{max} = 1023$ for a 10-bit converter.

In an NTC, heat lifts charge carriers across the band gap E_g . This gives the Arrhenius law $R(T) = R_0 \exp[\beta(1/T - 1/T_0)]$, with $\beta = E_g/2k_B$ and $T_0 = 298.15$ K. Dividing by R_f and taking the log gives a straight line:

$$\ln r = \frac{\beta}{T} + c, \quad c = \ln r_0 - \frac{\beta}{T_0}. \quad (3)$$

Let $A = 1/\beta$ and $B = -c/\beta$. Then solving for temperature gives the formula used in the logger:

$$T = \frac{1}{A \ln((1023 - ADC)/ADC) + B} - 273.15. \quad (4)$$

Table 5. Summary of the derivation.

Voltage divider	$r = (1023 - ADC)/ADC$; cancels V_{cc} noise.
Arrhenius	$R(T) \propto e^{\beta/T}$; carriers excited across E_g .
Linear form	$\ln r = \beta(1/T) + c$; the slope gives $\beta = 3965.3$ K.
Runtime	$T = 1/(A \ln r + B) - 273.15$; used in the logger.

β -Model Validation

We fitted a straight line to $\ln r$ against $1/T$ (Figure 3). It gives $\beta = 3965.3$ K, $R^2 = 0.9993$ and an RMSE of **0.598 °C**, using only two parameters (β and r_0).

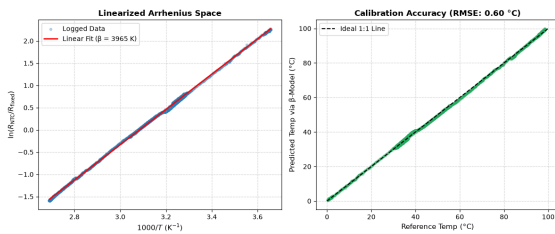


Figure 3. Straight-line Arrhenius fit (left) and β -model accuracy (right).

Response Time of the Calibrated NTC

Response-time curve
(calibrated NTC vs. DS18B20 reference)

Figure 4. Response-time curve of the calibrated NTC (to be added).

Time constant τ , rise time and method: to be added.

Sensitivity and Conclusion

The sensitivity $dT/dADC$ is 0.102–0.195 °C/count for the cubic and 0.101–0.245 °C/count for the β -model (Figure 5). The sensor reads best in the middle of the range and worse near both ends, where the divider output flattens.

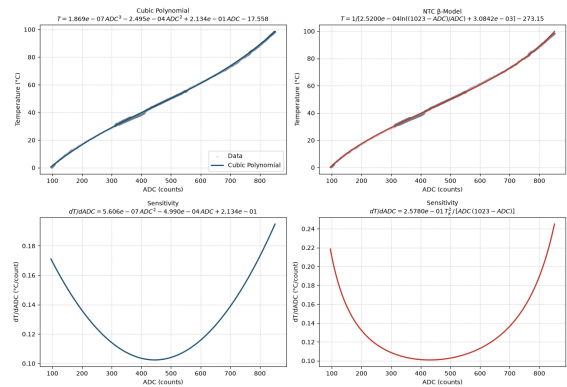


Figure 5. Fitted curves and sensitivity $dT/dADC$ for both models.

Table 6. Final model comparison.

Model	Params	RMSE	Notes
Global cubic	4	0.54 °C	Empirical; ± 1.06 °C (95 % PI)
NTC β -model	2	0.60 °C	Physical; $\beta = 3965$ K, $R^2 = 0.9993$

Both models calibrate the thermistor to within the ± 1 °C uncertainty of the DS18B20. Their RMSE differ by only 0.06 °C. The cubic is slightly better inside the measured range (0–98 °C). The β -model is almost as good with half the parameters, works well outside the range, and gives a real physical constant.

Limits of this work. (1) The reference thermometer is only accurate to ± 0.5 °C, and we assumed about ± 1 °C in practice, so errors below that cannot be proved. (2) The thermistor and the reference were not at exactly the same temperature during fast changes, which adds lag error. (3) Only three runs were usable, and the calibration covers 0–98 °C only. (4) The cubic should not be used outside this range, because polynomials can turn sharply beyond the data.

Recommendation. Use the β -model as the main calibration. Use the cubic only when you need the lowest error inside the measured range. The heating/cooling gap (about 0.6 °C on average) is smaller than the reference uncertainty, so one calibration works for both directions.